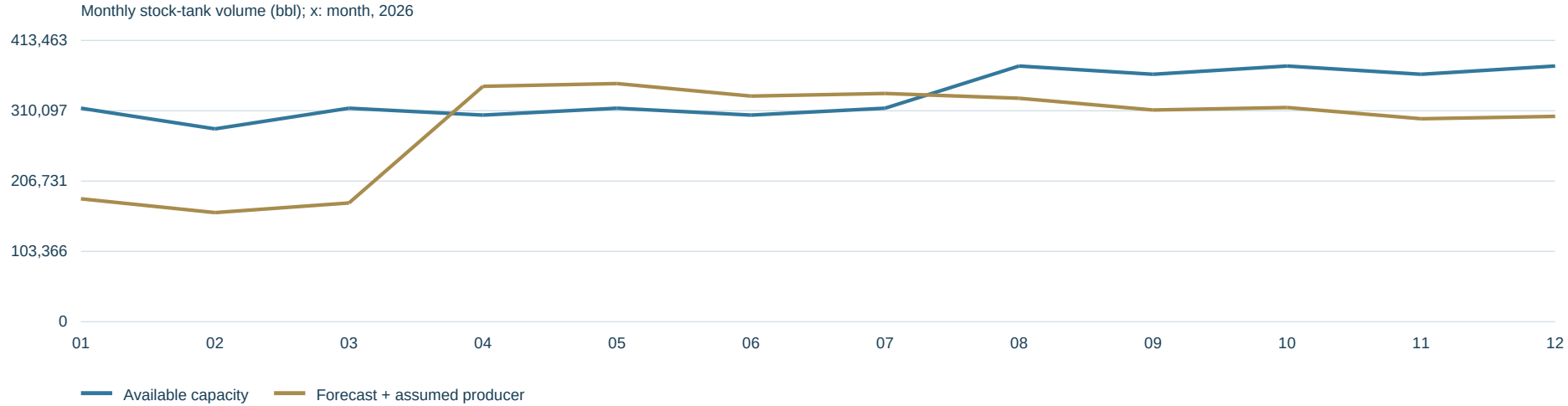
SWD water planning exhibit

Synthetic production and fictional assumed injectors; public reported evidence remains separate.

Frozen cutoff 2025-12. Producer starts 2026-04; available expansion 2,000 STB/d from 2026-08.

Capacity shortfall: 4 of 12 months. Annual unmet planned volume: 128,120 bbl. Peak planned demand: 11,536 STB/d.

Available capacity against planned demand



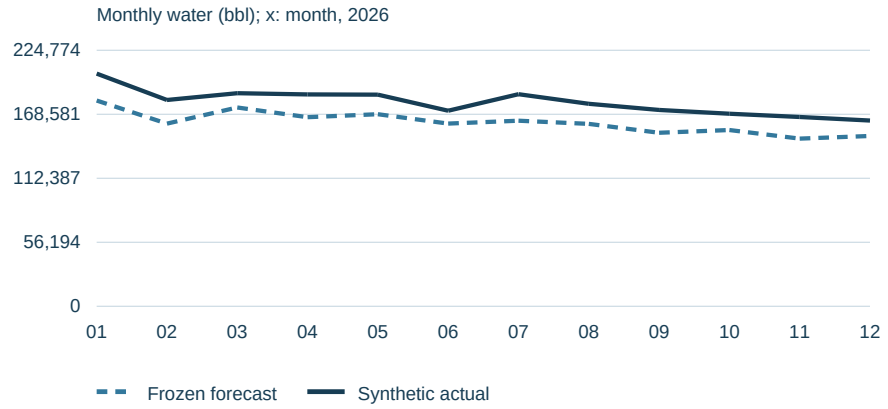
Assumed injector	Reservoir / limit psia	J STB/d/psi	Availability	Available STB/d
FICTIONAL-INJ-A	2,000 / 3,500	5	90%	6,750
FICTIONAL-INJ-B	2,000 / 3,500	2.5	90%	3,375

All engineering pressures are bottomhole psia at a common formation reference depth. Constant injectivity and pressure limits are assumed; no actual-well calibration or sustainable-capacity claim is made.

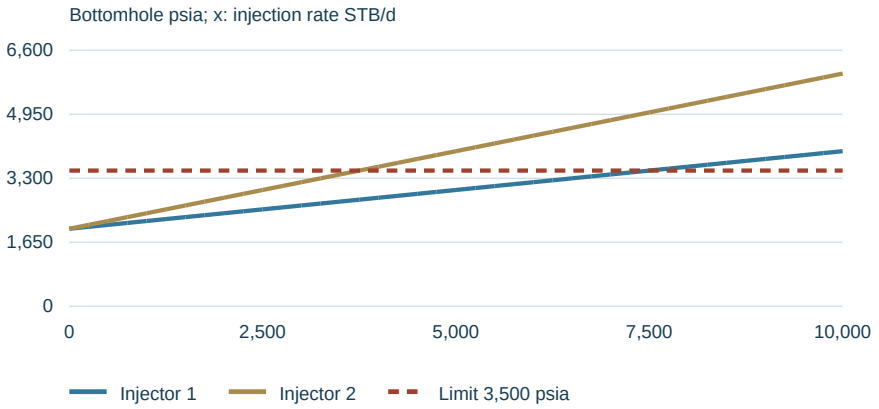
Reconciliation and injector response

Synthetic actual minus frozen forecast, original producers only; planned demand also includes the assumed new producer.

Frozen forecast / held-out actual



Pressure required to inject



Month	Days	Forecast bbl	Actual bbl	Variance %	Demand bbl	Capacity bbl	Headroom bbl
2026-01	31	180,698	204,340	13.1%	180,698	313,875	133,177
2026-02	28	160,425	181,183	12.9%	160,425	283,500	123,075
2026-03	31	174,582	187,127	7.2%	174,582	313,875	139,293
2026-04	30	166,069	186,060	12%	346,069	303,750	-42,319
2026-05	31	168,678	185,854	10.2%	350,086	313,875	-36,211
2026-06	30	160,455	171,748	7%	331,676	303,750	-27,926
2026-07	31	162,979	186,254	14.3%	335,539	313,875	-21,664
2026-08	31	160,203	177,828	11%	328,503	375,875	47,372
2026-09	30	152,396	172,425	13.1%	311,245	363,750	52,505
2026-10	31	154,796	169,109	9.2%	314,887	375,875	60,988
2026-11	30	147,254	166,249	12.9%	298,356	363,750	65,394
2026-12	31	149,575	163,125	9.1%	301,859	375,875	74,016

Pressure = reservoir pressure + rate / J. Dashed lines: assumed pressure limits; curve points above them are excluded rates. Injector numbers follow the first-page table. Calendar volumes use actual month lengths. Exact unrounded values are attached.

Public reported injection history

Public liquid injection is separate from synthetic production and fictional injector calculations.

UIC 000020838 / API 38910201 / well 5D; parent rrc:uic:givw-z9t4:2bb2e506dea36872:1

Artifact: rrc-2026-10-09-uic000020838-v1. 28 acquired source reports. Average and maximum pressure are separate reported psi fields; pressure basis is unspecified.

Month	Liquid bbl	Avg pressure psi	Max pressure psi	Calculated bbl/day	Source report ID
2024-01	17,173	900	900	553.97	3775425
2024-02	17,927	900	900	618.17	3775426
2024-03	17,989	900	900	580.29	3775427
2024-04	15,119	900	900	503.97	3775428
2024-05	22,427	900	900	723.45	3775429
2024-06	23,579	900	900	785.97	3775430
2024-07	24,010	900	900	774.52	3775431
2024-08	15,980	900	900	515.48	3775432
2024-09	15,820	900	900	527.33	3775433
2024-10	16,072	900	900	518.45	3775434
2024-11	16,108	900	900	536.93	3775435
2024-12	19,670	900	900	634.52	3775436
2025-01	22,695	900	900	732.1	3775437
2025-02	19,623	900	900	700.82	3775438
2025-03	20,142	900	900	649.74	3775439
2025-04	18,143	900	900	604.77	3775440
2025-05	20,487	900	900	660.87	3775441
2025-06	18,728	900	900	624.27	3775442
2025-07	14,727	900	900	475.06	3775443
2025-08	13,205	900	900	425.97	3775444
2025-09	9,283	900	900	309.43	3775445
2025-10	9,358	900	900	301.87	3775446
2025-11	9,096	900	900	303.2	3775447
2025-12	8,776	900	900	283.1	3775448
2026-01	8,134	900	900	262.39	3775449
2026-02	5,087	900	900	181.68	3775450
2026-03	8,041	900	900	259.39	3775451
2026-04	4,799	900	900	159.97	3775452

Not reported means missing, never zero. Calendar-day average = reported monthly liquid / calendar days. Monthly pressures do not establish bottomhole pressure or injectivity. Other inventory candidates have no acquired history here; no zero injection is inferred.

Method, assumptions and source lineage

The attached swd-technical-exhibit.json preserves the exact unrounded case, Unicode, public rows and full source metadata.

Technical exhibit SHA-256: 1e0e3dc6c0d615563660f46aed229b103e34931f047ff1ea9207ba62b968d6c1

Nexus local main de631ad2952bf42b80a6738ff980992975be4cdc. Fixture synthetic-water/seed-417/v1. Engineering constant-j-nexus-brine/1; forecast exponential-log-fit/1.

bottomholePressure: Pres + rate/J

pressureCapacity: max(0, J*(Plimit-Pres))

availableCapacity: min(pressureCapacity, operatingCeiling)*availability + explicitly assumed available expansion

monthlyVolume: calendar-day average rate * actual calendar days

variance: synthetic actual volume - frozen forecast volume

percentVariance: 100*variance/forecast; null when forecast=0

Assumptions and limitations

Forecast frozen at 2025-12; new-well schedule edits do not change actuals or fitted forecast.

Injectivity is fixed with pressure and time; no fracture, transient, layered or calibrated well model.

Assumed pressure limit is bottomhole psia, not a copied surface permit maximum.

Additional expansion is available stock-tank capacity after downtime; no implicit availability multiplier.

Nexus brine density uses temperature/salinity and ignores pressure; FVF/salinity and viscosity use simplified correlations and source clamps.

Public reported liquid injection is not assigned to synthetic producers or fictional injectors.

Calculated Nexus brine reference: 3000 psia, 180 F, 50000 ppm; Bw 1.0381 rb/STB; density 61.78 lb/ft3; viscosity 0.369 cP; compressibility 0.000002818378079782646 8 1/psi. ESP Nexus local main de631ad2; simplified brine correlations with source clamps; no calibration to a real well

Official source queries and preserved snapshots

inventory: RRC-UIC well location; retrieved 2026-10-09T11:11:47.999Z

https://data.texas.gov/resource/givw-z9t4.json?%24where=api_no+like+%27389%25%27+AND+uic_type_injection%3D%271%27+AND+permit_canceled_date+IS+NULL+AND+w3_plugged_date+IS+NULL&%24limit=5&%24order=uic_number

Snapshot SHA-256: 2bb2e506dea368726dcecfb5ad30f6a84fe3c5da8f3b43b0cf735a214396b45c

monthly: RRC-UIC H10 Injection Monitoring; retrieved 2026-10-09T11:11:48.192Z

https://data.texas.gov/resource/qq2j-f2zm.json?%24where=uic_no%3D%27000020838%27+AND+formatted_date+%3E%3D+%272024-01-01T00%3A00%3A00%27&%24limit=1000&%24order=formatted_date%2Cid

Snapshot SHA-256: de475d2906f07f5759d31627f0855bcaeb4145cb2752893e4dcf0b3b68ee5aed